

VISION FREE NAVIGATION-IOT POWERD SMART SHOE

COMMUNITY SERVICE PROJECT REPORT

Submitted by

A.JAYA CHARAN– 99220040237

K.VISHNU PREETHAM– 99220040276

T.HARSHASRI KARTHIKEYA– 99220040144

T. OM PRAKASH – 99220040206

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

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SCHOOL OF COMPUTING

COMPUTER SCIENCE AND ENGINEERING

KALASALINGAM ACADEMY OF RESEARCH

AND EDUCATION

KRISHNANKOIL 626 126

November 2024

DECLARATION

We affirm that the project work titled “**VISION FREE NAVIGATION-IOT POWERD SMART SHOE**” being submitted in partial fulfillment for the award of the degree of **Bachelor of Technology in Computer Science and Engineering** is the original work carried out by us. It has not formed part of any other project work submitted for the award of any degree or diploma, either in this or any other University.

A. Jayacharan

99220040237

K. Vishnu preetham

99220040276

T. Harshasri Karthikeya

99220040144

T. Om prakash

99220040206

This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

Date:

Signature of supervisor

Dr. R. Raja Subramanian

Associate Professor

Department of Computer Science and Engineering



KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
Under sec. 3 of UGC Act 1956. Accredited by NAAC with "A++" Grade



BONAFIDE CERTIFICATE

Certified that this project report “**VISION FREE NAVIGATION-IOT POWERD SMART SHOE**” is the bonafide work of “ **A.JAYACHARAN (99220040237), K.VISHNU PREETHAM (99220040276),T.HARASHASRIKARTHIKEYA(99220040144),T.OMPRAKESH(99220040206)** ” who carried out the project work under my supervision.

Dr. R. Raja Subramanian
SUPERVISOR

Associate Professor

Computer Science and Engineering
Kalasalingam Academy of Research and
Education
Krishnankoil 626126
Virudhunagar District.

Dr. N. Suresh Kumar
HEAD OF THE DEPARTMENT
Professor & Head

Computer Science and Engineering
Kalasalingam Academy of Research and
Education
Krishnankoil 626126
Virudhunagar District.

Submitted for the Project final review and Viva-voce held on

Internal Examiner

External Examiner

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SCHOOL OF COMPUTING
COMPUTER SCIENCE AND ENGINEERING
PROJECT SUMMARY

Project Title	Vision-free Navigation: lot-powered smart shoe	
Project Team Members (Name with Register No)	A. Jayacharan (99220040237) K. Vishnu preetham(99220040276) T. Harashasri Karthikeya(99220040144) T. Om Prakash(99220040206)	
Guide Name/Designation	Dr. R. Raja Subramanian/Associate Hod	
Program Concentration Area	IoT-powered assistive navigation, utilizing embedded systems, sensors, GPS, and energy-efficient technology for visually impaired individuals	
Technical Requirements	Arduino Uno, ultrasonic and IMU sensors, GPS module, haptic and audio feedback devices, rechargeable battery with piezoelectric energy harvesting, and supporting components for efficient operation and real-time navigation.	
Engineering standards and realistic constraints in these areas		
Area	Codes & Standards / Realistic Constraints	Tick ✓
Manufacturability	This constraint mainly occurs with hardware Manufacturing, Which integrates the entire system into one microcontroller, the hardware sensor must be adaptive for different sizes of shoes based on the requirements.	
Technical Constraints	There are many exist solutions, but the connection with parental control and Arduino Uno has limited computational capacity, making it challenging to run AI/ML algorithms locally.	

ABSTRACT

In today's fast-paced world, ensuring the mobility and safety of visually impaired individuals is a growing concern. Traditional mobility aids, such as walking canes and guide dogs, while helpful, often fall short in busy, dynamic, or unfamiliar environments. These limitations emphasize the need for an innovative and comprehensive solution. This study introduces an **Internet of Things (IoT)-powered smart shoe**, designed to empower visually impaired individuals with enhanced navigation, obstacle detection, and real-time tracking. The smart shoe leverages cutting-edge technology to offer a seamless and efficient experience, ensuring both safety and independence.

At the core of the system are **ultrasonic sensors** and **infrared (IR) sensors**, which together enable robust obstacle detection within a 4-meter radius. The ultrasonic sensors emit high-frequency sound waves that bounce back upon encountering an obstruction, allowing the shoe to calculate the distance and detect potential hazards in the user's path. This setup provides comprehensive front and side coverage, ensuring that users can navigate confidently in various environments. To further improve usability, the system includes an auditory buzzer and haptic feedback through vibration motors. When an obstacle is detected, these feedback mechanisms deliver instant alerts, enabling users to react swiftly and avoid potential dangers. The dual-alert system is especially valuable in noisy environments or for individuals with additional hearing impairments.

One of the standout features of the smart shoe is its **real-time location tracking** capability, powered by a GPS module. The GPS continuously tracks the user's position and transmits this information to an accessible interface. Family members or caregivers can monitor the user's movements and current location, ensuring their safety and offering peace of mind. This feature is particularly useful in crowded urban areas, where visually impaired individuals might face challenges navigating independently.

In conclusion, the IoT-powered smart shoe redefines how we address mobility and safety for visually impaired individuals. Its blend of advanced technology and practical features ensures a safer, more inclusive society. By empowering users with independence and reliability, this innovation paves the way for a future where wearable assistive technology becomes an indispensable tool for improving the quality of life for people with disabilities.

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LIST OF ABBREVIATION

IR - Infrared Sensor

US - Ultrasonic Sensor

GPS - Global Positioning System Module

IMU - Inertial Measurement Unit (MPU6050)

MCU - Microcontroller Unit (Arduino Uno)

Piezo - Piezoelectric Sensor

PWM - Pulse Width Modulation (used for haptic feedback control)

BZR - Buzzer

VM - Vibration Moto

CHAPTER –I

INTRODUCTION

Among the biggest obstacles that visually impaired people still confront are mobility and navigation. Even moving freely in familiar areas, negotiating a congested environment, or even crossing a busy street can be dangerous for millions of people globally and necessitate ongoing outside help. Recent estimates indicate that 2.2 billion individuals worldwide are affected by some kind of visual impairment. This comprises 258 million people with mild impairments, 295 million with moderate to severe impairments, and 43 million people who are totally blind. It is concerning to note that 90% of blind people live in low-income communities, where blindness is disproportionately common. The difficulties of being independent are made worse in these areas by the lack of access to resources like guide dogs, mobility aids, and cutting-edge technology.

Even though assistive technologies—like wearable's, GPS-enabled gadgets, and AI-powered solutions—have advanced significantly, many of these breakthroughs are still too costly, complicated, or unavailable to people in environments with limited resources. Despite being widely utilized, traditional aids like guide dogs and white canes are frequently insufficient in complicated or dynamic contexts, such as busy cities with constantly shifting barriers. Furthermore, there is a significant gap in the ability of visually impaired people to move independently and confidently because these solutions lack real-time feedback and comprehensive spatial awareness.

This study suggests a Smart Shoe Navigation System based on the Internet of Things (IoT) to overcome these constraints and give visually impaired people more mobility and safety. In order to provide real-time obstacle identification, navigation assistance, and location monitoring, the system combines a number of cutting-edge technologies, such as GPS module, ultrasonic sensors, and infrared (IR) sensors. The device uses an auditory buzzer and haptic feedback (vibration) to notify the user when it detects an obstruction, allowing for prompt and efficient reactions. In order to ensure sustainable operation and lessen the need for frequent recharging, the smart shoe also uses piezoelectric technology, which transforms the user's bodily movements into electrical energy.

This Internet of Things-enabled smart shoe offers a socially beneficial solution in addition to a technological advancement. It tackles the particular mobility issues experienced by visually

impaired people in low-resource environments by fusing cost with useful functionality. Users can roam freely and safely in both familiar and unfamiliar surroundings because to the system's increased autonomy. Incorporating a GPS module also adds a degree of security and peace of mind by enabling family members and caregivers to track the user's location in real time.

This project's main objective is to create a navigation aid that is impactful, accessible, and easy to use by bridging the gap between cost and cutting-edge technology. By encouraging independence, safety, and confidence in their everyday mobility, the Internet of Things-powered smart shoe is intended to improve the quality of life for those with visual impairments. This study shows how technology may change lives and solve important issues that underprivileged communities confront by utilizing contemporary IoT technologies.

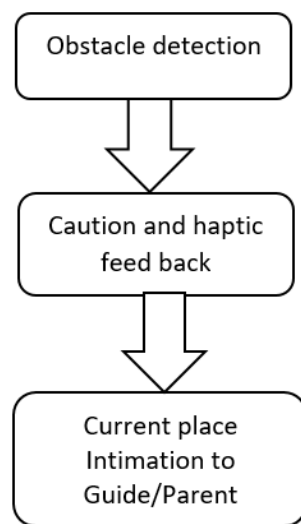


Figure 1: Architecture of System

CHAPTER-II

LITERATURE REVIEW

By using sensors to identify elevation changes and offer haptic feedback, the smart insole shoe helps blind people navigate stairs and obstacles. Stability on stairs is ensured via a heel-locking mechanism. Design that prioritizes safety and aims to increase the freedom of visually impaired people is informed by modeling, surveys, and patent research. By identifying impediments and providing navigational guidance, the "Smart Shoes" technology helps people with visual impairments. An Arduino receives data from embedded sensors and uses it to connect to a smartphone app for auditory notifications. The app encourages independence and safe mobility by integrating with Google Maps to deliver instructions and obstacle notifications. By employing ultrasonic sensors to identify obstructions at the knee, ground, and side levels, the "Smart Shoe System" helps those with vision impairments. It is controlled by an Arduino microcontroller and allows autonomous navigation by providing audio feedback through headphones. GPS integration for destination navigation is one possible future improvement. A research team created smart shoes with GPS navigation and 89.5% accurate obstacle recognition based on Arduino for the blind and visually handicapped. The system provides vibrations and audio-visual alerts via sensors, Bluetooth, and a smartphone app. Although it has limits when it comes to recognizing small things, testing revealed an average response time of 3.08 seconds. A research team created smart shoes with GPS navigation and 89.5% accurate obstacle recognition based on Arduino for the blind and visually handicapped. The system provides vibrations and audio-visual alerts via sensors, Bluetooth, and a smartphone app.

CHAPTER III

PROBLEM DEFINITION AND PROJECT OBJECTIVES

Navigating their surroundings is extremely difficult for visually impaired people, especially in busy or new settings. The effectiveness of traditional mobility aids, such as guiding dogs and white canes, is restricted because they frequently don't offer early warning signs of obstacles or need a large financial commitment and ongoing upkeep. Visually impaired people become less confident and independent as a result of becoming dependent on outside help. Additionally, the absence of real-time tracking and communication capabilities makes it difficult for caregivers to guarantee these people's safety. In low-income communities, where access to sophisticated navigation systems is restricted, the issue is more severe and many people lack workable answers. The lack of accessible, effective, and user-friendly mobility aids emphasizes how urgently a novel approach that improves safety is needed

Create an Obstacle Detection System in Real Time

- Combine ultrasonic and infrared (IR) sensors to identify impediments up to four meters away and deliver prompt feedback via auditory alarms and haptic massages.
- Include Real-Time Location Monitoring, By using a GPS module, family members and caretakers may keep an eye on the user's whereabouts and remotely guarantee their safety.
- Utilize Sustainable Solutions to Increase Energy Efficiency,Reduce your need on external power sources by utilizing piezoelectric sensors to transform physical activity into electrical energy.
- Create a Solution That Is Both Accessible and Affordable, Provide a smart shoe system that is affordable, lightweight, and easy to use for those living in low-resource environments.
- Encourage self-reliance and self-assurance in mobility, Give visually impaired people a dependable navigation tool so they may move around freely and safely in a variety of settings.

CHAPTER IV

PROPOSED METHODOLOGY

IoT-powered Navigation System, The Arduino Uno acts as the main control unit in vision-free navigation, coordinating the actions of all connected parts. It controls essential features like real-time location updates, user caution and haptic feedback, and obstacle detection. The system is in line with internationally accepted standards for IoT-based assistive technologies thanks to the incorporation of IEEE standards, which guarantee interoperability, data correctness, and system stability

Identifying Obstacles

The obstacle detection system uses four 4-meter-range ultrasonic sensors, two of which are built into each shoe. Sound waves at 40 kHz are emitted by these sensors, bounce off of obstructions, and then return to the sensor. The distance to the obstacle is determined by measuring the time it takes for the sound waves to return. This data is processed by the Arduino Uno to efficiently identify barriers.

Redundancy and spatial awareness are ensured by the configuration's extensive front and side coverage. This helps the system identify hazards like steps, curbs, or other surrounding obstructions in the walking route, which is especially useful in low-visibility situations. The sensors guarantee accurate and reliable performance for real-time obstacle detection by conforming to IEEE 1451.4 standards

System of User Intimation

To provide real-time alerts, the user intimation system combines buzzers and vibration motors. The Arduino Uno initiates two feedback methods in response to the ultrasonic sensors detecting an obstacle

Haptic Feedback: A tactile alert is produced by activating the vibration motors in the shoe closest to the obstruction.

Aural Feedback: To alert the user, the buzzer simultaneously emits an audible alert. This dual-feedback technology ensures that even those with hearing impairments can avoid obstacles because it works well in both noisy and quiet surroundings. The system complies with IEEE 802.15.4 standards, guaranteeing dependable sensor-to-Arduin connection for prompt feedback

Integration of GPS Modules

The Neo-6M GPS Module is used by the system to deliver real-time location updates. When combined with the Arduino Uno, this small and energy-efficient GPS module provides accurate latitude and longitude data at predetermined intervals.

Live location tracking is made possible by the module's transmission of data in NMEA (National Marine Electronics Association) format to the Arduino's serial monitor. With the use of this tool, mentors or caregivers can keep an eye on the user's whereabouts via the interface. High precision in time synchronization and position data is ensured by using IEEE 1588 standards, which are essential for dependable GPS operation

IMU Sensor (MPU6050):

We have selected the MPU6050 IMU Sensor, which combines both an accelerometer and gyroscope, enabling accurate motion and orientation detection. This sensor connects to the microcontroller with the help of the I2C protocol, facilitating efficient data transfer. So by using this sensor, we are finding step count, calories burned, and distance travelled.

Step Count:

To detect steps, the MPU6050 acceleration values in X, Y, and Z axes is read. The accelerometer data is converted from raw values to “g” units by dividing by 16384.0. The magnitude of the acceleration vector MMM is calculated as follows:

- $M = \sqrt{x^2 + y^2 + z^2}$
- where x, y, and z are the accelerations along each axis.

Distance Calculation:

The step length is set to an average of 0.762 meters. The total distance traveled is calculated by multiplying the step count by the step length:

- Total Distance=Step Count×Step Length

This method allows for real-time tracking of the total distance covered by the user.

Calorie Burn Calculation:

To estimate the calories burned, a calorie factor is used. This factor represents the average calories burned per step, set to 0.05 calories per step. The calories burned are calculated as:

- $\text{Calories Burned} = \text{Step Count} \times \text{Calorie Factor}$

The implementation of IEEE standards, such as IEEE 802.3 (Ethernet) for data communication and IEEE 21451 (smart transducer interface) for sensor integration, ensures that the IMU sensor operates reliably and efficiently.

- IEEE 1451.4 for calibration and accuracy of sensors.
- IEEE 802.15.4 for low-power and secure sensor-controller connection.
- For accurate time synchronization in GPS data, use IEEE 1588.
- IEEE 21451 is used to standardize interfaces for smart transducers.

By following these guidelines, the IoT-powered navigation system making it a strong option for helping those with visual impairments.

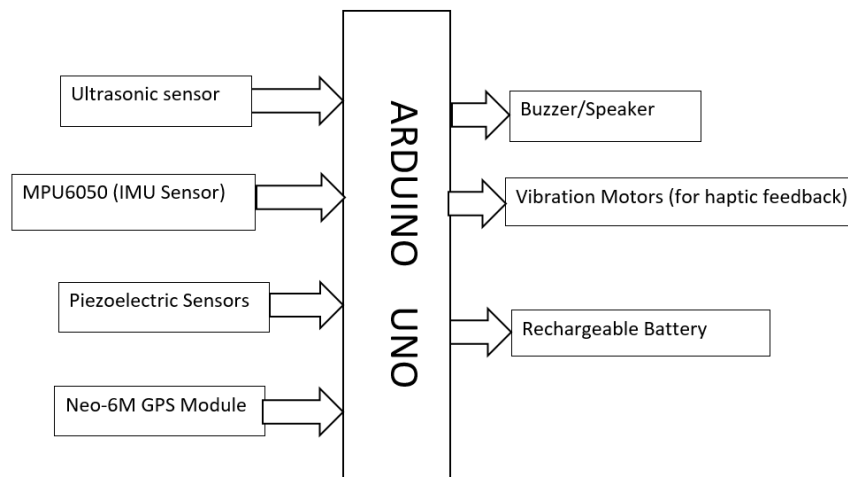


Figure 2 :Block diagram, Aurdino uno is mother of this system, the input are from IMU sensor, obstacle detection Ultra sonic sensor, Piezoelectric Sensor for regenerating energy, Gps module for live location detection. Buzzer and vibration motor as output for quick cautioning.

CHAPTER V

REQUIREMENTS AND MODULE DESCRIPTION

Hardware Requirements

- **Arduino Uno:** Acts as the central microcontroller to manage and integrate all components.
- **Ultrasonic Sensors (HC-SR04):** Detect obstacles up to 4 meters and provide front and side coverage.
- **Infrared (IR) Sensors:** Enhance obstacle detection in short-range scenarios.
- **Neo-6M GPS Module:** Provides real-time location data for tracking and user assistance.
- **MPU6050 IMU Sensor:** Combines accelerometer and gyroscope to calculate step count, distance, and calorie burn.
- **Vibration Motors:** Deliver haptic feedback to alert users about obstacles.
- **Buzzer/Speaker:** Provides audio feedback for immediate warnings.
- **Rechargeable 3.7V Li-ion Battery:** Powers the system with the potential to recharge through energy harvesting.
- **Piezoelectric Sensors:** Convert walking energy into electrical energy to recharge the battery.

Software Requirements

Arduino IDE: For programming the Arduino Uno microcontroller.

Python: For post-processing and interfacing with tracking applications.

Google Maps API: For live location tracking and visualization.

Module Description

Obstacle Detection Module

Components: Ultrasonic sensors and Arduino Uno.

Functionality: Uses ultrasonic waves to detect obstacles within a range of 4 meters. The data is processed to calculate the distance between the user and the object, ensuring safety in both static and dynamic environments.

User Intimation Module

Components: Vibration motors, buzzers, and Arduino Uno.

Functionality: Alerts the user through haptic (vibrations) and audio feedback when an obstacle is detected. The alert intensity varies based on proximity, with stronger vibrations for closer objects.

GPS Tracking and Live Location Module

Components: Neo-6M GPS Module and Google Maps API.

Functionality: Provides real-time location data to the user and their caregivers. The module sends latitude and longitude values periodically, enabling precise tracking through a user interface.

Motion Analysis Module

Components: MPU6050 IMU Sensor.

Functionality: Tracks step count, calories burned, and distance traveled using accelerometer and gyroscope data.

Step Count: Detects steps by analyzing changes in acceleration along the X, Y, and Z axes.

Distance Calculation: Multiplies step count by the average step length to determine the total distance traveled.

Calorie Burn Estimation: Uses a predefined calorie factor to calculate energy expenditure.

Power Management Module

Components: Rechargeable Li-ion battery, piezoelectric sensors, and voltage regulator.

Functionality:

Converts walking-induced energy into electrical energy using piezoelectric sensors.

Regulates power to ensure stable operation of all components.

CHAPTER VI

SYSTEM IMPLEMENTATION

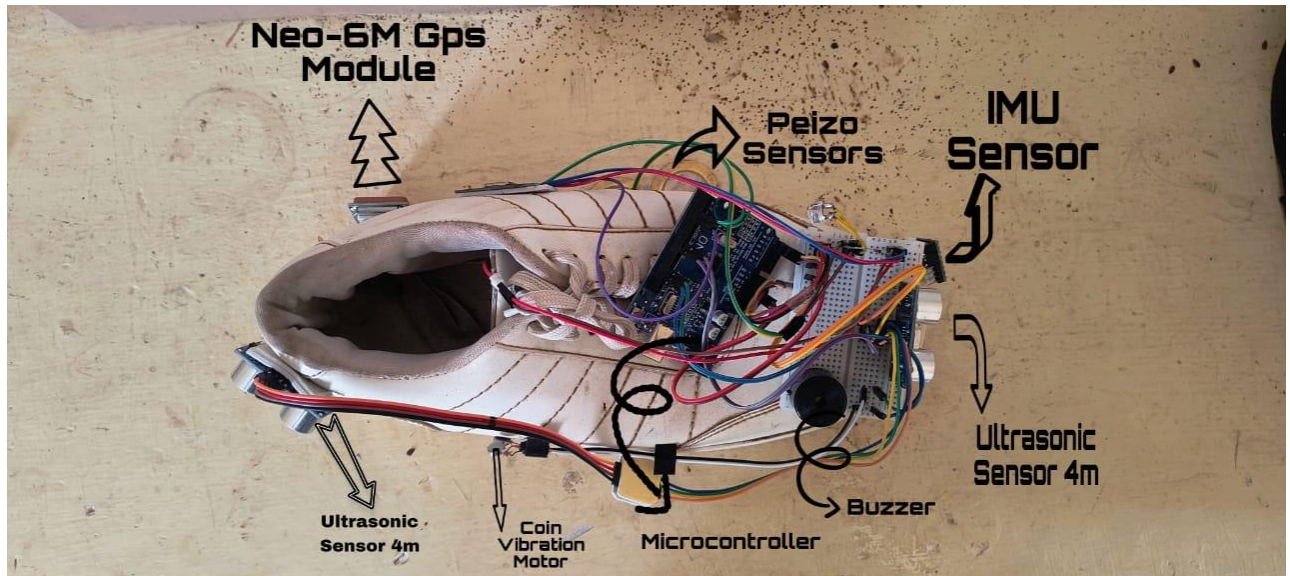


Figure 3: System working

The IoT-powered navigation system is designed to assist users, particularly those with visual impairments, by providing real-time obstacle detection and feedback. The system is built around an Arduino, which serves as the central control unit. Ultrasonic sensors detect obstacles by emitting sound waves and measuring the time it takes for the echo to return. When an obstacle is identified, the Arduino processes the sensor data and activates a buzzer and vibration motor. This dual feedback system ensures the user receives both auditory and haptic alerts, making it suitable for various environments and users, including those with hearing impairments.

The system is powered by a rechargeable lithium-ion battery, which is enhanced with an energy-harvesting mechanism. Piezoelectric sensors embedded in the system generate energy from the user's movements, charging the battery as they walk. This self-sustaining design reduces reliance on external charging and ensures long-term usability.

The combination of obstacle detection, immediate feedback, and energy efficiency makes the system a practical and reliable assistive tool. By leveraging renewable energy and innovative feedback methods, it offers enhanced mobility and safety for users, while maintaining compactness and efficiency, making it ideal for everyday use.

CHAPTER VII

RESULTS AND DISCUSSION

RESULT

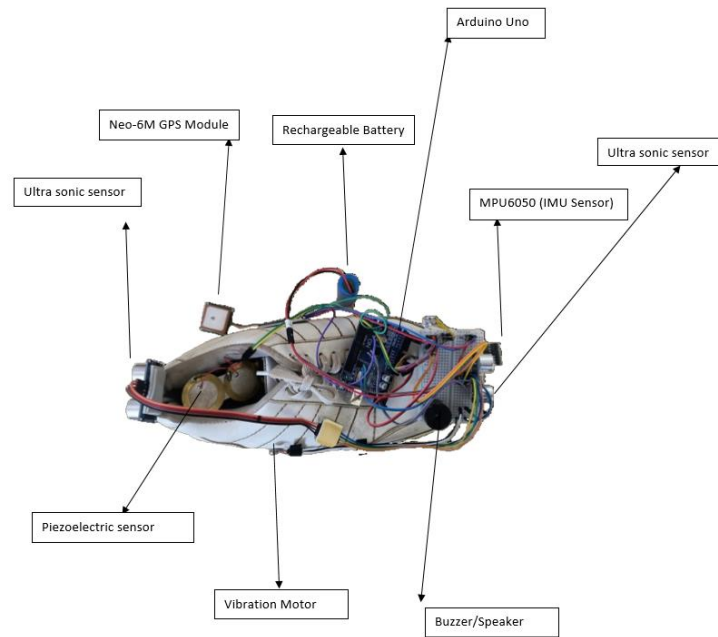


Figure 4: Overall System Implementation

DISCUSSION

By combining GPS for real-time tracking, dual feedback mechanisms that use sound and vibration, and ultrasonic sensors for obstacle identification, the Internet of Things-powered vision-free navigation system improves mobility and safety for people with visual impairments. By encouraging self-reliance and self-assurance, the approach successfully lessens dependency on caretakers. An environmentally friendly touch is provided via sustainable energy harvesting using piezoelectric sensors; nonetheless, more power is required for complete operation. False positives in detection are a minor problem that can be solved with better algorithms. This creative approach is a step ahead in assistive technology, providing future chances for intelligent developments by promoting accessibility, autonomy, and physical well-being.

CHAPTER VIII

COMMUNITY IMPACT

Visually impaired people and the general public are expected to be significantly impacted by the IoT-powered vision-free navigation system that uses smart sneakers. This initiative intends to empower visually impaired people to go through daily life with greater confidence by improving their mobility, freedom, and safety. With the use of haptic feedback, auditory alarms, and obstacle detection, users may steer clear of any dangers and move more safely in a variety of settings. In addition to enhancing their quality of life, this independence lessens their dependency on family members and caregivers, which lessens the mental and physical strain on those who offer support. Furthermore, by enabling visually challenged people to engage more completely in social interactions, the technology fosters inclusivity.

Additionally, by giving visually impaired people the resources they need to travel securely and confidently, the project promotes their involvement in the labor, which benefits the economy. Because of its affordability, the technology will continue to be available to those in low-income communities, where eye impairment is more common. Additionally, it encourages innovation in assistive technology, which may result in fresh approaches to other accessibility issues. Since the project makes use of energy-efficient technology, such as the Neo-6M GPS module, and may include sustainable manufacturing practices and energy harvesting features, its environmental impact is similarly significant. Additionally, by raising awareness and advocating for the needs of the blind and visually impaired, the project can promote more inclusive public space designs and legislation. The project can broaden its reach and guarantee that more people take advantage of this transformative technology by cultivating alliances with NGOs, academic institutions, and governmental organizations. All things considered, the IoT-powered navigation system has the ability to bring about long-lasting social change by giving visually impaired people more autonomy and fostering a more accessible and inclusive society for everybody

CHAPTER IX

CONCLUSION AND FUTURE SCOPE

With its sophisticated infrared and ultrasonic sensors, the Internet of Things-powered smart shoe offers a revolutionary solution for people with vision impairments by increasing their mobility and guaranteeing their safety in a variety of settings. These sensors provide prompt alerts to avert possible mishaps by detecting obstructions up to 4 meters away. Real-time location tracking is made possible by the incorporation of a GPS module, enabling family members or caretakers to keep an eye on the user's whereabouts and guarantee their wellbeing. This feature adds an extra degree of security and is particularly useful in crowded or unfamiliar places.

Additionally, the shoe uses cutting-edge energy-harvesting technology that uses piezoelectric sensors to transform walking-related calories burned into useful energy. By reducing the need for regular charge, this self-recharging feature makes the device incredibly practical and environmentally friendly. Even in noisy environments, the shoe's dual feedback mechanisms—haptic vibrations and aural alerts—ensure that users are instantly aware of any obstructions.

The smart shoe is a dependable, easy-to-use assistive technology that combines energy efficiency, location monitoring, and obstacle recognition. It increases self-confidence, lessens need on outside help, and enables visually impaired people to navigate on their own. This cutting-edge technology greatly enhances users' quality of life and is a prime example of how the Internet of Things may be used to develop useful assistive technology

Future improvements to the IoT-powered smart shoe for the blind and visually impaired are highly promising. To enhance navigation, sophisticated AI algorithms can be incorporated for predictive path analysis and obstacle classification. Voice-guided directions might be made possible by real-time Bluetooth or Wi-Fi connectivity with cellphones, which would further improve the system's use. Using LiDAR technology could enable 3D mapping and more accurate distance estimations. An additional layer of usefulness might be added by the addition of health monitoring sensors, which would detect vital indications like oxygen levels and heart rate. Future iterations may be lighter and smaller, making the shoe more comfortable and portable while increasing the energy harvesting efficiency to support more features.

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Paper Title

Vision-Free Navigation:lot-Powerd Smart Shoe

Abstract

There are many obstacles to overcome in order to ensure the safety and mobility of people with visual impairments, which calls for ongoing study and creative solutions. Identifying and avoiding obstacles in one's path is one of the main challenges that blind persons confront. In order to overcome these obstacles, this article presents an Internet of Things (IoT)-powered smart sneaker that combines the latest developments with focused on users features. To identify obstacles within a 4-meter radius, the system makes use of essential Internet of Things components, such as ultrasonic and infrared (IR) sensors. Real-time obstacle recognition is made possible by these sensors, particularly in overcrowded or dangerous environments like roadways. When the shoe detects an obstruction, it instantly makes a buzzing sound and vibrates to provide haptic feedback, allowing the user to react quickly. The smart shoe is also outfitted with ped with a GPS module, enabling tracking of location in real time. This function improves the user's safety and gives their loved ones peace of mind by enabling guardians or family members to keep an eye on their whereabouts via an intuitive interface. The energy produced when walking is transformed into electrical energy by the shoe's modern piezoelectric technology. This lessens reliance on external charging solutions in addition to enabling self-recharging. The system also monitors activity levels and steps, which helps to improve physical health by revealing patterns of movement. This all-inclusive solution provides a dependable mobility aid for visually impaired people in a variety of settings, bridging the gap between safety, self-determination, and sustainability.

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Authors

Jaya Ambati (kalasalingam Univarsity) <ambatijayacharan18@gmail.com> ✓



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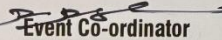
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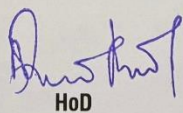
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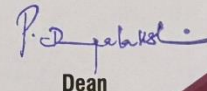
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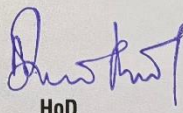
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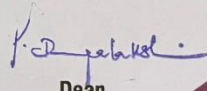
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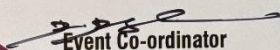
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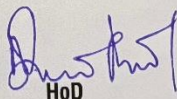
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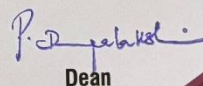
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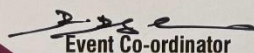
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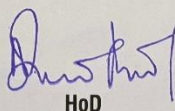
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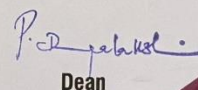
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PROJECT AUDIT REPORT

This is to certify that the project work entitled “**VISION FREE NAVIGATION-IOT POWERD SMART SHOE**” categorized as an internal project done by **A. JAYACHARAN, K.VISHNU PREETHAM T.HARSHASRI KARTHIKEYA,T.OMPRAKASH** of the Department of Computer Science and Engineering, under the guidance of **Dr.R.Raja Subramanian** during the Odd semester of the academic year 2023 - 2024 are as per the quality guidelines specified by IQAC.

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